HEALTH INSURANCE COST ANALYSIS REPORT

SQL Data Extraction | Power BI Dashboard | Business Intelligence Analysis

Prepared by: Dayo Rex Afariogun

Role: Data Analyst

Tools Used: SQL • Power BI • Microsoft Excel

Project Type: Healthcare Insurance Cost Analysis

Executive Summary

This project analyses healthcare insurance charges using demographic and lifestyle information extracted from a SQL database and visualized in Power BI.

The objective was to determine the major factors influencing insurance charges while identifying patterns that can help insurers improve pricing strategies, manage financial risk and develop preventive healthcare initiatives.

The analysis shows that smoking status is the strongest predictor of insurance cost, while BMI and age significantly increase medical expenses. Gender and geographical region have relatively little influence compared to lifestyle-related risk factors.

Overall, the findings demonstrate that behavioural factors have a greater financial impact than demographic characteristics.

Project Objectives

The objectives of this project were to:
*Extract insurance data using SQL
*Clean and validate the dataset
*Build an interactive Power BI dashboard
*Analyse factors influencing insurance charges
*Identify high-risk customer groups
*Generate business recommendations supported by data

Data Overview

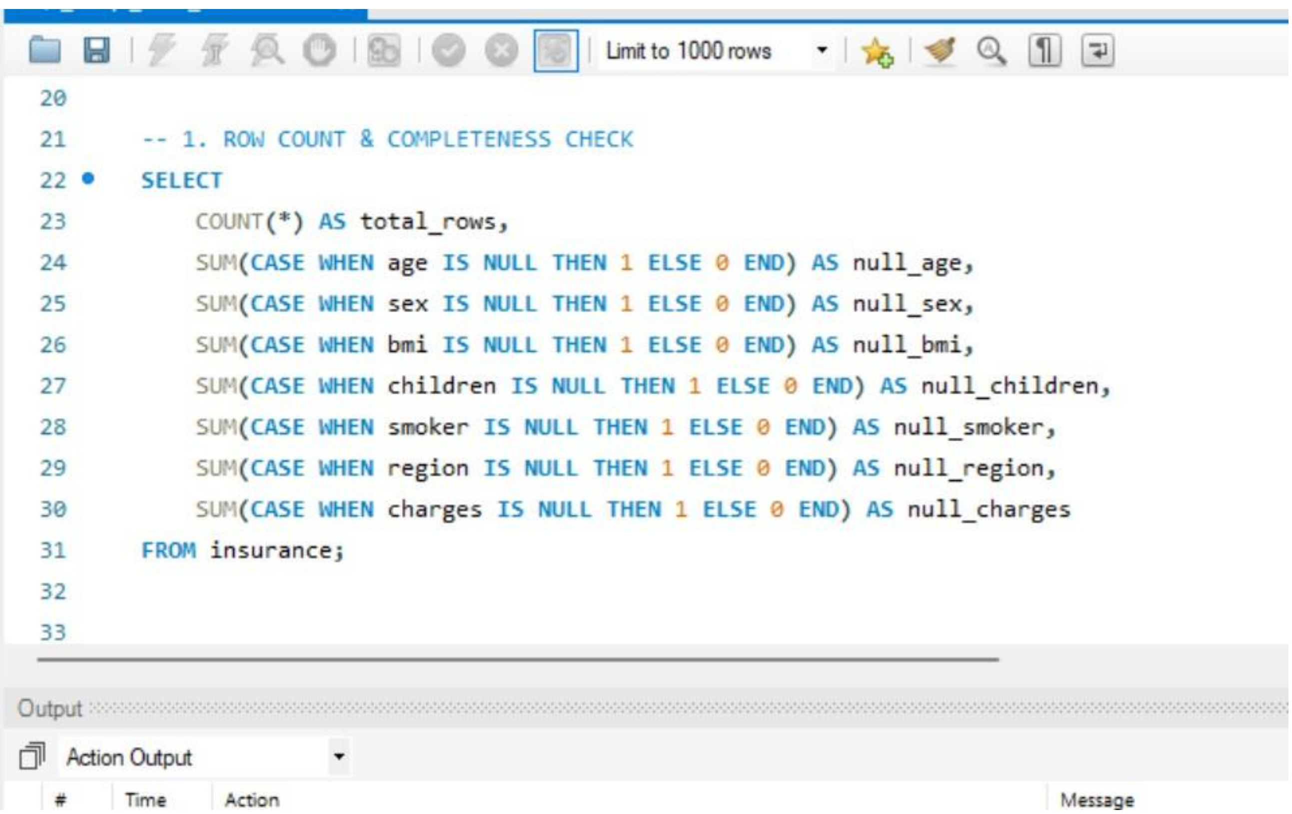
The dataset contains customer demographic and health-related information including:
*Age
*Sex
*BMI
*Number of Children
*Smoking Status
*Region
*Medical Charges
These variables represent common factors used by insurance companies when assessing customer risk and determining premium prices.

SQL Data Extraction

The project began by retrieving the required fields from the insurance database.
Example SQL query:

```
SELECT
Age,
Sex,
BMI,
Children,
Smoker,
Region,
Charges
FROM Insurance;
```

Additional SQL techniques included:
*Filtering records
*Sorting data
*Basic aggregation
*Preparing clean data for Power BI



Limit to 1000 rows

```sql
-- DATA PREP FOR VISUALIZATION
-- Table: analyst_portfolio.insurance
-- Output: Clean, aggregated views for Dashboard
-- ============================================
USE analyst_portfolio;

-- 1. CREATE CLEANED BASE VIEW
-- Calculated fields added in order not to repeat logic in PowerBI
CREATE OR REPLACE VIEW vw_insurance_clean AS
SELECT
    age,
    sex,
    bmi,
    children,
```

Output

Action Output

#	Time	Action	Message

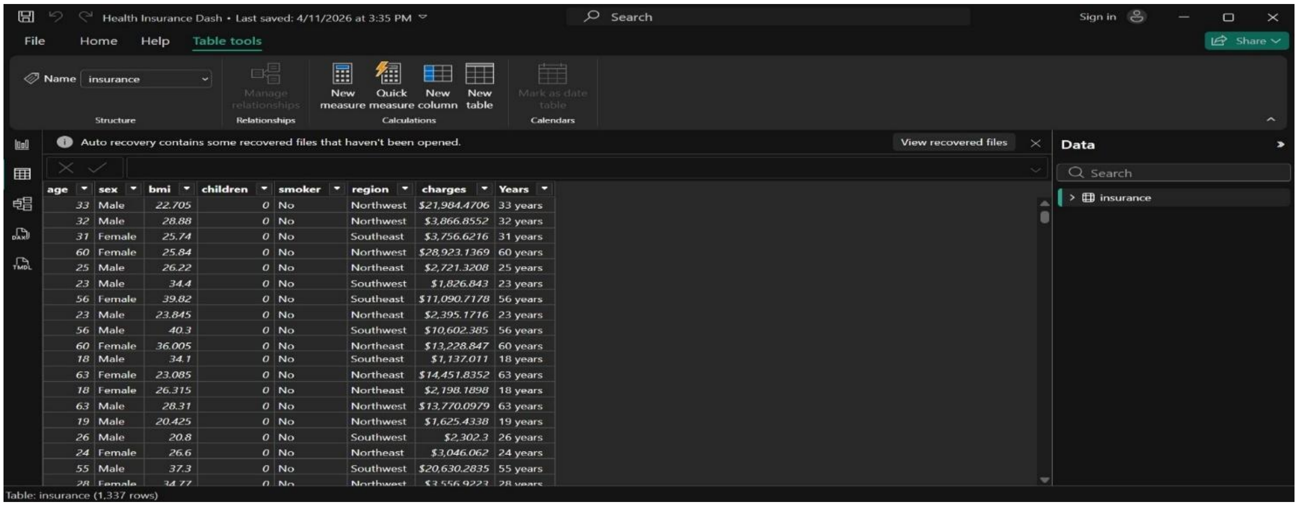
Limit to 1000 rows

```sql
            WHEN age < 30 THEN '18-29'
            WHEN age BETWEEN 30 AND 45 THEN '30-45'
            WHEN age BETWEEN 46 AND 60 THEN '46-60'
            ELSE '60+'
        END AS age_group,
        CASE
            WHEN bmi < 18.5 THEN 'Underweight'
            WHEN bmi BETWEEN 18.5 AND 24.9 THEN 'Normal'
            WHEN bmi BETWEEN 25 AND 29.9 THEN 'Overweight'
            ELSE 'Obese'
        END AS bmi_category,
        CASE WHEN smoker = 'yes' THEN 'Smoker' ELSE 'Non-Smoker' END AS smoker_status
    FROM insurance
    WHERE charges > 0 AND bmi > 0 AND age > 0;  -- Basic DQ filter
```

Output

Action Output

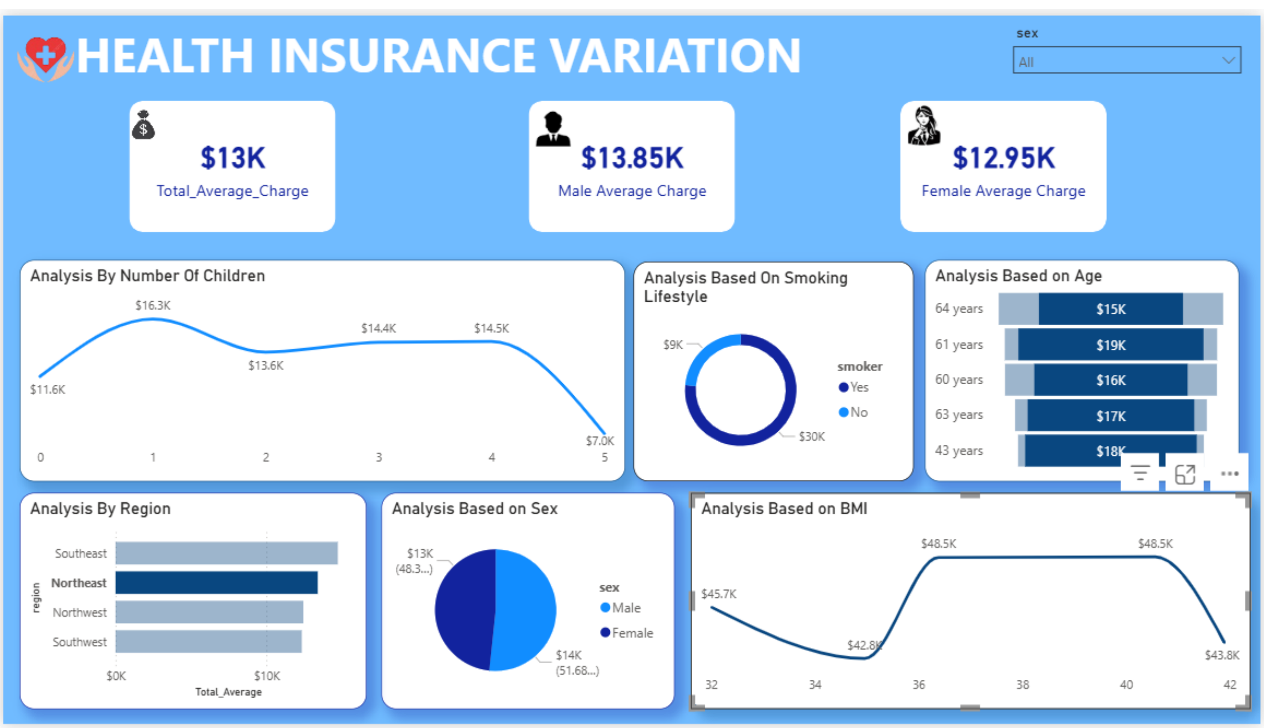
#	Time	Action	Message



Dashboard Overview

The Power BI dashboard provides an interactive overview of insurance costs using KPI cards, slicers and visualizations.

Dashboard components include:
*Overall Average Insurance Charge
*Average Male Charges
*Average Female Charges
*Charges by Smoking Status
*Charges by Age
*Charges by Region
*Charges by BMI
*Charges by Number of Children
*Interactive Gender Filter



The dashboard enables decision-makers to quickly identify the major drivers of insurance costs.

Key Performance Indicators (KPIs)

KPI	Value
Overall Average Charge	$13,000
Male Average Charge	$13,850
Female Average Charge	$12,950

Interpretation
Male customers have only a slightly higher average insurance cost than females.
The difference is relatively small, indicating that gender alone is not a major pricing factor.

Detailed Business Analysis

1. Smoking Status is the Largest Cost Driver
The dashboard clearly shows the biggest gap exists between smokers and non-smokers.
Smokers incur average healthcare costs approximately three times higher than non-smokers.

Business Interpretation
Smoking increases the likelihood of:
*Heart disease
*Lung cancer
*Stroke
*Chronic respiratory illnesses
*Long-term medical treatment

This leads to:
*Higher claim frequency
*Larger claim amounts
*Greater financial risk for insurers
Smoking therefore becomes one of the most heavily weighted variables in insurance pricing.

2. Age Significantly Increases Healthcare Costs
Insurance charges increase steadily with age.
Older individuals naturally require more healthcare services due to increased health risks.

Business Impact
Older customers typically generate:

*More medical consultations
*More prescription medication
*More hospital admissions
*Higher insurance claims

Insurance companies therefore increase premiums gradually as customers age.

3. BMI Strongly Influences Insurance Charges
The dashboard shows that customers with higher BMI values experience significantly higher insurance costs.
This relationship becomes even stronger when combined with smoking.

Business Interpretation

High BMI increases the likelihood of:
*Diabetes
*Hypertension
*Heart disease
*Joint replacement
*Kidney disease

Customers with both obesity and smoking habits represent the highest financial risk.

4. Number of Children
Insurance charges increase moderately as the number of dependants increases.
However, the increase is not as dramatic as smoking or BMI.

Business Explanation
Families generally require:
*More healthcare services
*Child medical care
*Vaccinations
*Maternity services
Although additional dependants increase healthcare utilisation, lifestyle factors remain stronger predictors of cost.
5. Regional Analysis
The Southeast region shows slightly higher average charges than other regions.
However, regional differences are relatively small.

Business Interpretation
This suggests pricing should focus more on customer health characteristics rather than location.
Regional pricing alone would not effectively predict future healthcare costs.

6. Gender Analysis
Average charges between males and females are very similar.

Business Interpretation
Gender has minimal predictive power once variables such as:
*Smoking
*BMI
*Age
are considered.

This demonstrates that behavioural factors have a greater influence on healthcare expenditure than gender.

Financial and Business Impact
From an accounting perspective, these findings directly affect both the Profit & Loss Statement and the Balance Sheet of an insurance company.

Revenue
Insurance premiums collected from customers represent revenue.

Higher-risk customers generally pay higher premiums.
However, if pricing is inaccurate, the company may collect insufficient premium income to cover future claims.

Claims Expense
Smoking and obesity significantly increase claim costs.
If claims grow faster than premium income:
*Operating profit decreases
*Underwriting profit declines
*Net income reduces

Profitability
Healthy customers improve profitability because:
*They pay premiums
*They make fewer claims
This increases the company's underwriting margin.

Balance Sheet Impact
The analysis also affects several Balance Sheet accounts.

Cash
Higher claim payments reduce cash reserves.

Insurance Liabilities
Insurance companies maintain claim reserves for expected future payouts.
Higher-risk customer groups require larger reserves.
This increases liabilities.

Retained Earnings
If claim expenses become excessive:
*Net profit decreases
*Retained earnings reduce
*Shareholders' equity declines

Investments
Insurance companies invest premium income before claims are paid.
Lower claim costs allow more funds to remain invested, generating additional investment income.

Strategic Business Recommendations
Recommendation 1
Develop wellness programmes encouraging:
*Smoking cessation
*Weight management
*Preventive healthcare
This reduces future claim costs.

Recommendation 2
Introduce risk-based pricing models using:
*Age
*BMI

*Smoking Status
rather than relying heavily on demographic variables.

Recommendation 3
Reward healthy customers with premium discounts.
This encourages healthier behaviour while improving customer retention.

Recommendation 4
Use predictive analytics and machine learning to identify high-risk policyholders early.
This enables proactive intervention before expensive claims occur.

Recommendation 5
Focus preventive healthcare campaigns on customers with both high BMI and smoking habits, as this combination creates the greatest financial risk.

Power BI Skills Demonstrated

This project demonstrates practical experience with:
*SQL Data Extraction
*Data Cleaning
*Data Modelling
*Power Query
*Interactive Dashboards

*KPI Cards
*DAX Measures
*Slicers and Filters
*Data Storytelling
*Business Intelligence Reporting

Key Business Insights

*Smoking is the strongest predictor of insurance cost.
*BMI significantly increases healthcare expenditure.
*Older customers generate higher medical costs.
*Gender has minimal impact after controlling for lifestyle factors.
*Regional differences are relatively small.
*Preventive healthcare programmes could substantially reduce long-term insurance costs.
*Risk-based pricing supported by analytics can improve profitability while maintaining fair premiums.

Conclusion

This project demonstrates how SQL and Power BI can be used to transform raw healthcare data into meaningful business intelligence. Rather than simply visualizing insurance charges, the analysis identifies the underlying drivers of cost and translates them into actionable recommendations for pricing, risk management, and operational strategy.

The results show that lifestyle choices—particularly smoking and obesity—have a far greater influence on insurance costs than gender or region. These insights enable insurers to make more informed underwriting decisions, strengthen financial performance, and promote preventive healthcare initiatives.

From a portfolio perspective, this project showcases my ability to combine SQL, Power BI, analytical thinking, and business knowledge to solve real-world problems. It demonstrates not only technical proficiency in data extraction and visualization, but also the ability to communicate insights that support executive decision-making. These are core competencies expected of a Data Analyst in healthcare, insurance, financial services, and other data-driven organisations across Ireland.